Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18IOT7DE52]

# SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

## DATASCIENCE FOR IOT

**Time: 3 Hours** **Maximum Marks: 70**

**Instructions:**
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

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| **1.** | **a.** | With a neat block diagram explain the features of Massive Parallel Processing Architecture. | **07** |
| | **b.** | List the features of Big Data Platform. | **07** |
| **2.** | **a.** | Differentiate between Univariate, Bivariate Data. Find the Standard Deviation, Variance, mean, median, mode and Range for the following Sample Data Set. (7,11,11,15,20,20,28) | **07** |
| | **b.** | Explain the various steps involved in the process of Dimensionality Reduction using PCA. | **07** |
| **3.** | **a.** | With a neat diagram explain the two phases involved in the Map Reduce programming model. | **07** |
| | **b.** | Perform the Matrix vector Multiplication operation using Map Reduce Algorithm for the following matrices. A= 1 2 / 3 4 B= 3 4 / 2 1 | **07** |
| | | **OR** | |
| **4.** | **a.** | Explain Mapreduce parallel dataflow | **10** |
| | **b.** | List and Explain components of MapReduce | **04** |
| **5.** | **a.** | What is Apache Pig? List and explain its features. | **07** |
| | **b.** | Explain How Apache Sqoop Works. | **07** |
| | | **OR** | |
| **6.** | **a.** | Differentiate between SQL and NoSQL Databases? | **07** |
| | **b.** | Explain the various steps involved in the process of Data Analysis using R. | **07** |
| **7.** | **a.** | Explain the concept of filtering streams with examples. | **07** |
| | **b.** | Discuss the concept of decaying window in detail. | **07** |
| | | **OR** | |
| **8.** | **a.** | With a neat sketch explain the architecture of data-stream management system. | **07** |
| | **b.** | Write a short note on sampling in Data Streams. | **07** |